

Distances Occurring Between One and n Times

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Abstract

Let $A \subset \mathbb{R}^2$ be a set of n points. We study distances determined by A whose multiplicity lies between 1 and n . The literal statement that there must always be two such distances is false, because an equilateral triangle determines only one distance. On the other hand, a standard theorem on the largest distances in planar point sets implies that, for all sufficiently large n , there are at least two such distances, and in fact the number of such distances tends to infinity with n .

1 Statement of the problem

For a finite set $A \subset \mathbb{R}^2$, let

$$\Delta(A) := \{|x - y| : x, y \in A, x \neq y\}$$

be the set of distinct distances determined by A . For $d \in \Delta(A)$, write $m_A(d)$ for the number of unordered pairs $\{x, y\} \subset A$ with $|x - y| = d$.

We are interested in distances $d \in \Delta(A)$ satisfying

$$1 \leq m_A(d) \leq n.$$

Define

$$L(A) := \#\{d \in \Delta(A) : 1 \leq m_A(d) \leq n\}.$$

For each n , define the extremal quantity

$$g(n) := \min\{L(A) : A \subset \mathbb{R}^2, |A| = n\}.$$

The original questions are:

1. Must $g(n) \geq 2$?
2. Must $g(n) \rightarrow \infty$ as $n \rightarrow \infty$?

2 A small counterexample

Theorem 1. *The assertion $g(n) \geq 2$ is false in general.*

Proof. Take $n = 3$ and let A be the vertex set of an equilateral triangle. Then $\Delta(A)$ has exactly one element, namely the side length of the triangle, and that distance occurs exactly three times. Hence $L(A) = 1$, so $g(3) = 1 < 2$. $\square$

Thus the first question has a negative answer if interpreted literally for every n .

3 The asymptotic answer

The positive asymptotic statement follows from the following standard theorem on large distances.

Theorem 2 (Vesztergombi). *For every fixed integer $k \geq 1$, there exists $N(k)$ such that the following holds for all $n \geq N(k)$. If $A \subset \mathbb{R}^2$ has $|A| = n$ and if*

$$\delta_1 > \delta_2 > \cdots > \delta_s$$

are the distinct distances determined by A in decreasing order, then each of the distances $\delta_1, \dots, \delta_k$ occurs at most n times; that is,

$$m_A(\delta_i) \leq n \quad (1 \leq i \leq k).$$

Remark 3. *Each δ_i occurs at least once by definition, so the conclusion of the theorem is exactly that the k largest distinct distances all contribute to $L(A)$.*

Corollary 4. *There exists N_0 such that for every $n \geq N_0$ and every n -point set $A \subset \mathbb{R}^2$, one has $L(A) \geq 2$.*

Proof. Apply Vesztergombi's theorem with $k = 2$. For all $n \geq N(2)$, the two largest distinct distances δ_1 and δ_2 both satisfy

$$1 \leq m_A(\delta_i) \leq n.$$

Hence both are counted by $L(A)$, so $L(A) \geq 2$. Therefore $g(n) \geq 2$ for all sufficiently large n . $\square$

Corollary 5. *One has $g(n) \rightarrow \infty$ as $n \rightarrow \infty$.*

Proof. Let $M \geq 1$ be fixed. Apply Vesztergombi's theorem with $k = M$. Then there exists $N(M)$ such that whenever $n \geq N(M)$ and $A \subset \mathbb{R}^2$ has $|A| = n$, the M largest distinct distances

$$\delta_1, \dots, \delta_M$$

all satisfy

$$1 \leq m_A(\delta_i) \leq n.$$

Consequently $L(A) \geq M$ for every such A , and therefore

$$g(n) \geq M \quad (n \geq N(M)).$$

Since M is arbitrary, this proves that $g(n) \rightarrow \infty$ as $n \rightarrow \infty$. $\square$

4 Conclusion

The problem has the following exact resolution.

Theorem 6. *Let $g(n)$ be defined as above.*

1. *The inequality $g(n) \geq 2$ is false for all n , since $g(3) = 1$.*
2. *The inequality $g(n) \geq 2$ holds for all sufficiently large n .*
3. *In fact $g(n) \rightarrow \infty$ as $n \rightarrow \infty$.*